

Supplementary Material for BurnRiSc

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I. PROJECT STATISTICS

To situate our findings without compromising the privacy of the repositories and developers studied, we report descriptive statistics for the ten open-source projects included in the initial BurnRiSc validation set. We selected ten repositories to span a range of project profiles; for each, we summarize its domain, governance, popularity, age, and community size. Reporting these characteristics allows readers to gauge the diversity and representativeness of the sample. Each repository number corresponds to the developer number in the signals table in Section 5.1.

TABLE I: Descriptive statistics for the ten repositories in the BurnRiSc validation set. Values for Stars, Forks, Watchers, and Total Contributors are approximate.

Project ID	Domain	Governance	Age (yrs)	Stars	Forks	Watchers	Total Contributors
P1	Developer tooling	Distributed governance	14	2,500	450	50	250
P2	Data infrastructure	Foundation-hosted	15	17,600	1,500	250	380
P3	Developer tooling	Foundation-stewarded	13	61,300	7,200	810	300
P4	Systems software	Foundation-governed	15	116,000	15,400	1550	6,800
P5	Engineering / CAD software	Volunteer-run	11	4,100	600	100	100
P6	Developer tooling	Corporate-stewarded	5	1,600	260	30	40
P7	Systems software	Minimal-maintainance	11	36,100	2,300	250	359
P8	Developer tooling	Single-maintainer	13	25,500	1,700	250	140
P9	Accessibility documentation	Community-Run	13	3,900	600	190	230
P10	Developer tooling	Organization Maintained	14	15,400	2,700	410	310

As shown in Table I, the ten repositories span roughly two orders of magnitude in popularity (1.6k–116k stars) and a comparably wide range of domains and governance models, indicating that the validation set is not narrowly concentrated around one project type.

II. OPTIMIZED WEIGHTS

In RQ2, we discussed the average weights of the top three and bottom two metrics under the optimized weighting scheme. For completeness, we report the full weight vector for every metric under each of the four anchors evaluated. Table II lists the per-metric weight assigned under each profile; a fifth, unweighted *baseline* profile (all weights fixed at 1.0) is included in Figure 1 for comparison but omitted from the table since it carries no per-metric variation.

TABLE II: Per-metric weights after optimization with each anchor.

Metric	Equal	Heuristic	Disengagement Heavy	Exhaustion Heavy
Off-hours Activity	1.251	1.387	5.000	3.033
Lexical Arousal	0.525	0.651	0.050	1.273
Average Merge Time	0.062	0.594	0.050	0.330
Lexical Exhaustion Features	1.000	1.100	1.000	2.194
Sentiment Neutralization	1.000	0.900	1.000	1.796
Commit Volume	1.470	1.536	5.000	1.658
Review Volume	0.050	0.217	0.050	0.050
PR Rejection Rate	0.050	0.192	0.050	0.050
Task Abandonment	0.050	0.523	0.050	0.138
First-person Pronoun Use	1.000	0.900	1.000	0.900
Disengagement Lexical Features	1.000	1.100	1.000	1.100
Lexical Diversity	1.000	0.900	1.000	0.900
Hapax Rate	0.050	0.050	1.000	0.050
Sentiment Lowering	1.000	0.900	1.000	0.900

As Table II shows, all four profiles consistently downweight Review Rate, PR Rejection Rate, and Average Merge Time relative to the other signals, while Off-hours activity and Commit Volume receive some of the largest weights across.

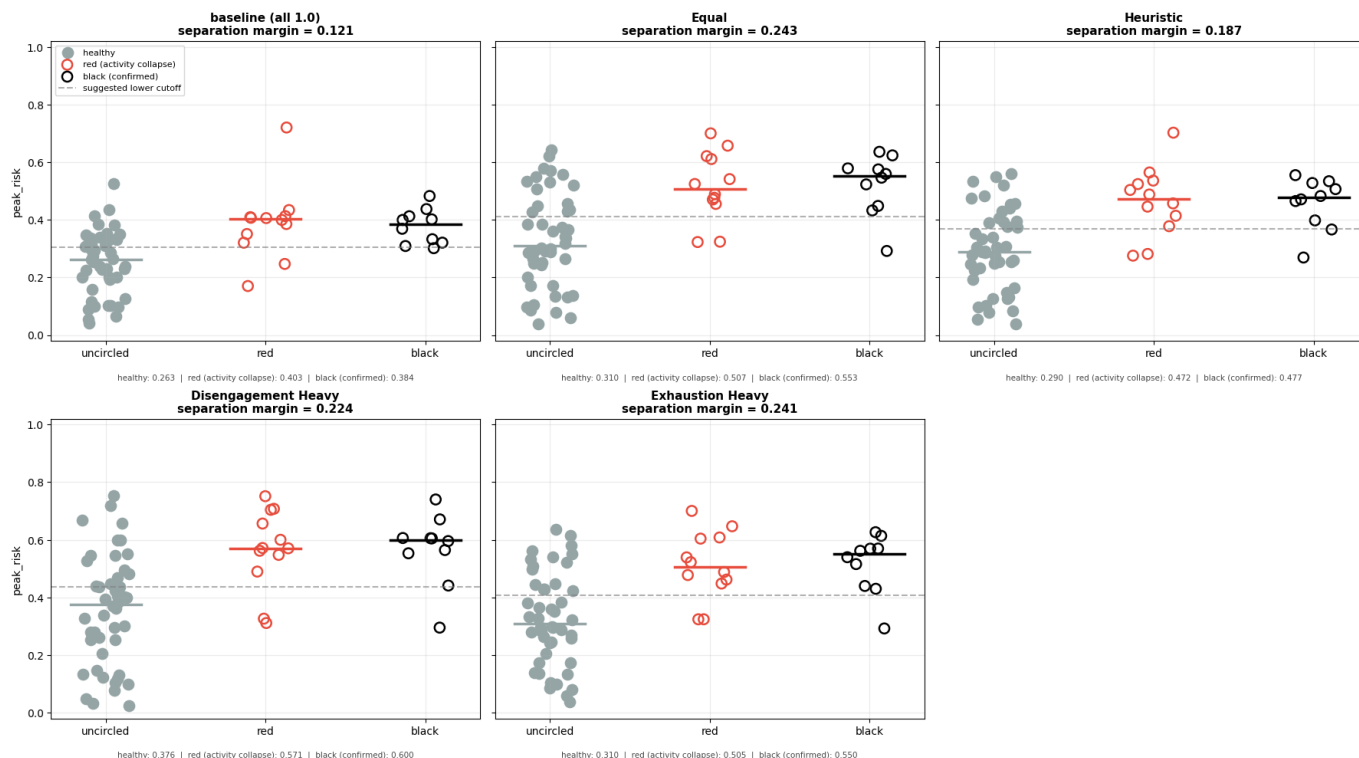


Fig. 1: Distribution of peak burnout-risk score (peak_risk) for healthy, activity-collapse (red), and confirmed (black). Horizontal bars mark group medians; the dashed line marks the suggested threshold. Separation margin (the gap between the healthy median and the lower of the two at-risk group medians) is highest under the *Equal* (0.243) and *Exhaustion Heavy* (0.241) anchors and lowest under the unoptimized *baseline* (0.121), indicating that optimization meaningfully improves separation between healthy and at-risk developers regardless of which profile is chosen.